	ITER, SIKSHA 'O' ANUSANDHAN (Deemed to be University)		Assignment 2
Branch	**Computer Science Engineering**	Programme	**B. Tech**
Course Name	Renewable Energy	Semester	**8TH**
Course Code	**OPL4001**	Academic Year	**2025-26**

Assignment- 2		**Topic:** Renewable Energy Principles	
Learning Level (LL)	**L1**: Remembering	**L3:** Applying	**L5**: Evaluating
	L2: Understanding	**L4**: Analysing	**L6**: Creating

Q's	Questions	COs	LL
1	In the context of PV define cell, module and array?	**CO3**	**L2**
2	Explain the photovoltaic principle and the PV system for electricity generation.	**CO3**	**L2**
3	Define solar constant? Express solar constant in terms of $MJ/m^2/h$.	**CO2**	**L2**
4	What is concentration ratio? How its value affects the efficiency of collectors, explain?	**CO2**	**L2**
5	Name the instrument used for measuring global radiation. Explain the construction and working of the instrument.	**CO2**	**L2**
6	Describe the principle of solar photovoltaic energy conversion	**CO3**	**L2**
7	What do you understand by energy payback period	**CO3**	**L2**
8	With the help of block diagrams explain the operations of stand-alone and grid Interactive SPV systems.	**CO3**	**L2**
9	Describe the classification of solar cells based on the type of active material used	**CO3**	**L2**
10	Give a neat diagram of a central tower receiver power plant and explain its operation.	**CO3**	**L2**

Note:

1. Assignment carries a weightage of 10 **marks out of 100**
2. CO2 and CO3 were covered.

Course Outcomes	CO1	Understand different sources of energy and their effects in developing aware ness on impacts of global warming.
	CO2	Solar Thermal energy and Understanding how Solar radiation reaches the earth's surface.
	CO3	Solar Photo Voltaic system, its type and its integration.
	CO4	Hydro energy, types of hydroelectric plant its environmental impact.
	CO5	Tidal power, Power generation from tidal barrages and its environmental impact.
	CO6	Wind energy, different types of wind turbine. Power and energy from wind turbine and its environmental impact.